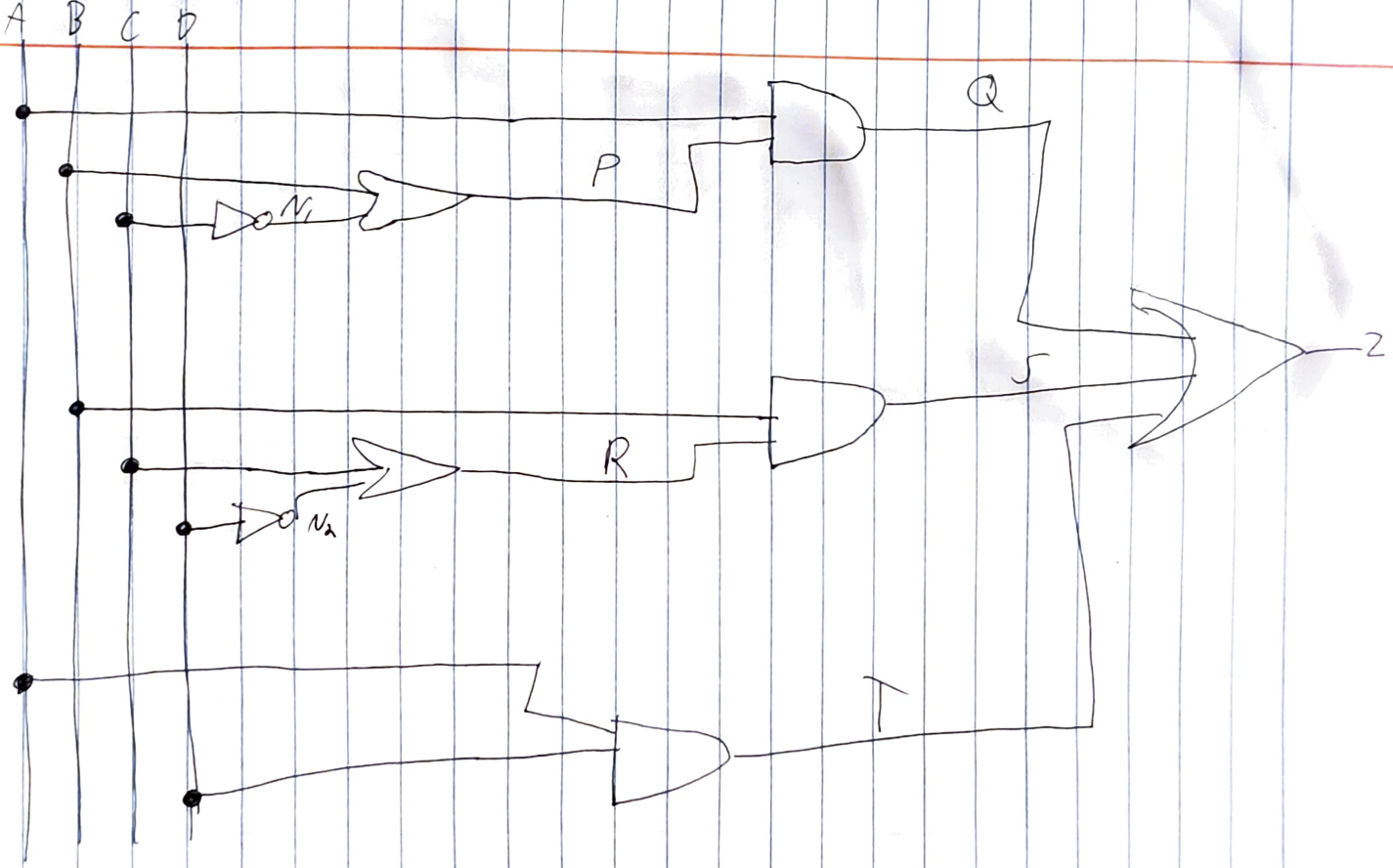
Q2 A



Q2 B

$N_1 = \bar{C}$

$N_2 = \bar{D}$

$P = B + N_1 = B + \bar{C}$

$Q = A \cdot P = A(B + \bar{C})$

$R = C + N_2 = C + \bar{D}$

$S = B \cdot R = B(C + \bar{D})$

$T = AD$

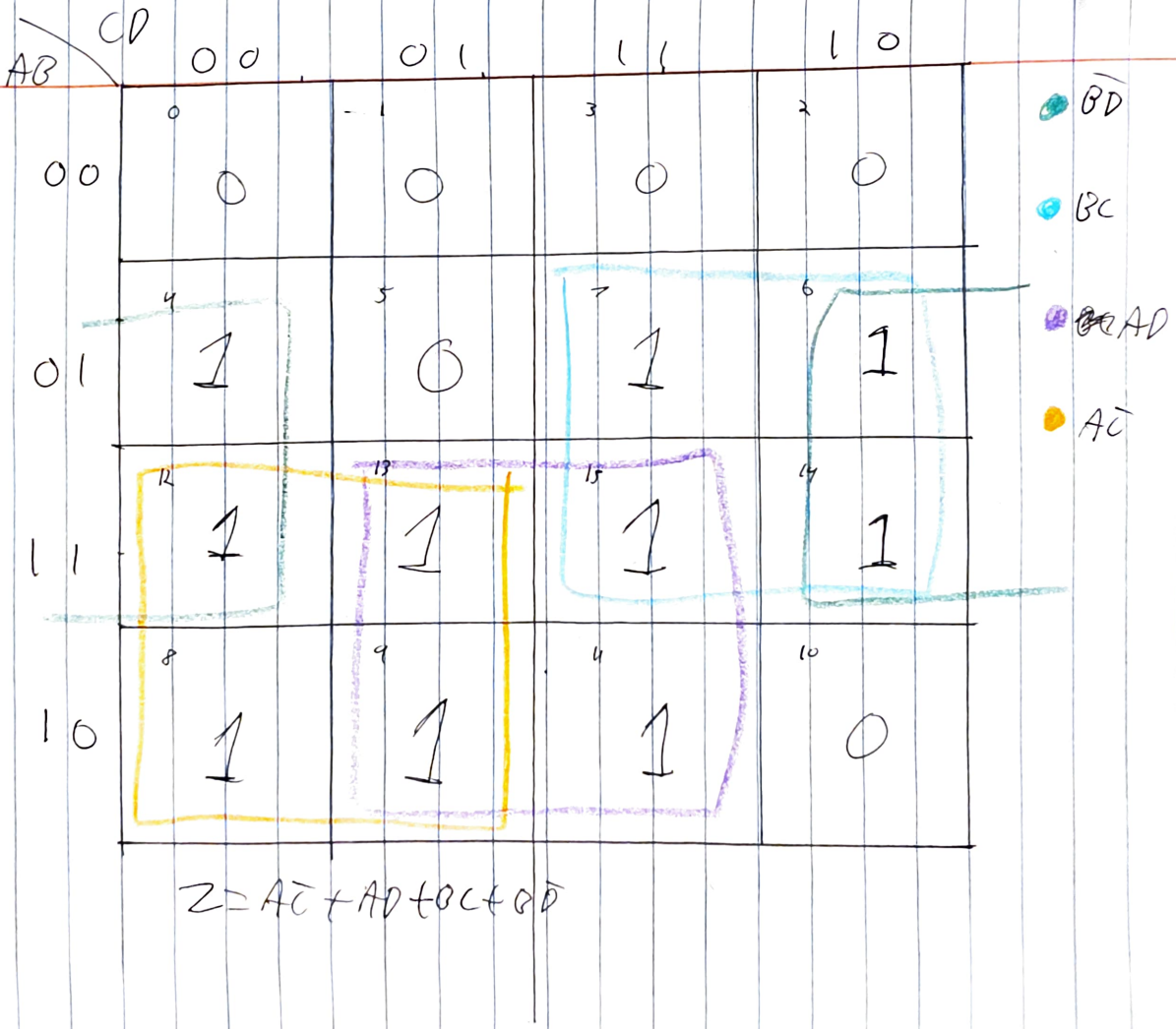
$Z = Q + S + T = A(B + \bar{C}) + B(C + \bar{D}) + AD$

$$Z = AB + A\bar{C} + BC + B\bar{D} + AD$$

(C)

A	B	C	D	N_1	N_2	P	Q	R	S	T	Z
0	0	0	0	1	1	0	1	0	1	0	0	0
0	0	0	1	1	0	1	0	0	0	0	0
0	0	1	0	0	1	0	0	1	0	0	0
0	0	1	1	0	0	0	0	1	0	0	0
0	1	0	0	1	1	1	1	1	1	0	1
0	1	0	1	1	0	1	1	0	1	0	0
0	1	1	0	0	1	0	0	1	1	0	1
0	1	1	1	0	0	0	0	1	1	0	1
1	0	0	0	1	1	1	1	1	1	0	1
1	0	0	1	1	0	1	1	0	1	1	1
1	0	1	0	0	1	0	0	1	0	0	0
1	0	1	1	0	0	0	0	1	0	1	1
1	1	0	0	1	1	1	1	1	1	0	1
1	1	0	1	1	0	1	1	0	0	1	1
1	1	1	0	0	1	0	1	1	1	0	1
1	1	1	1	0	0	0	1	1	1	1	1

Q2 D



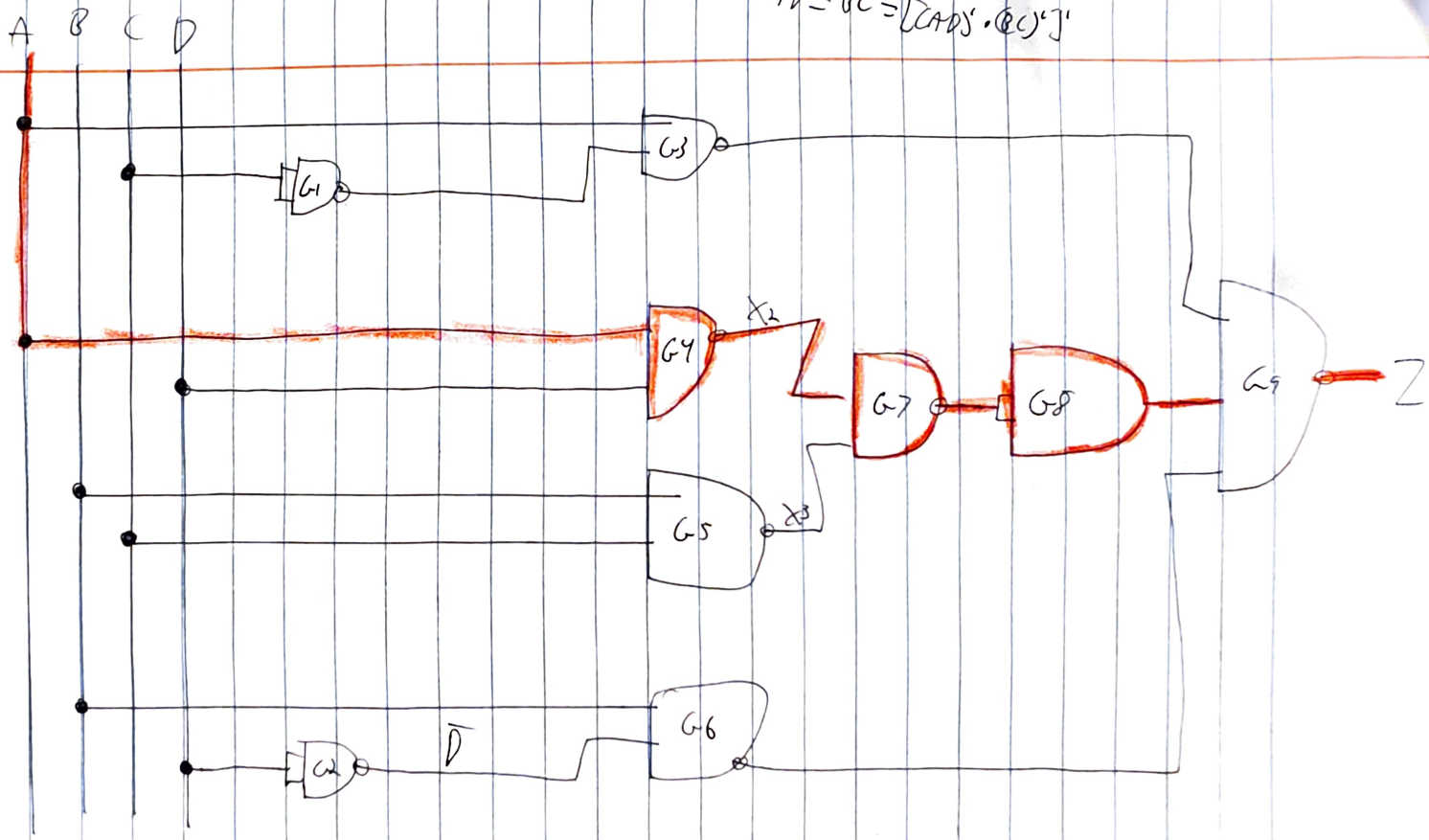
$Z = A\bar{C} + A\bar{D} + B\bar{C} + B\bar{D}$

Q2 E

$Z = A\bar{C} + B\bar{D} + (AD + BC)$

$Z = [(A\bar{C})' \cdot (B\bar{D})' \cdot (AD + BC)']'$

$AD = BC = [(AD)' \cdot (BC)']'$



(F) Longest Path (Red) → A → G4 → G7 → G8 → G9 → Z

Worst case delay → 4 gates • 20ns = 80ns